Student Enrollment & Course Management System

(SECMS)

A Salesforce Application for Academic Enrollment Management

Project Documentation

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1. Introduction

1.1 Project Overview

This project aims to centralize all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access, structured workflows, and accurate reporting. The system allows Enrollment Officers to manage student admissions, course registrations, and communication effectively, while instructors and administrators gain full transparency into course allocations and academic progress. The solution also enables automated instructor assignment, task creation, validation checks, and multi-level approval processes for enrollment management.

With Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization can ensure that all enrollment operations — from admission to course completion — are handled systematically and with continuous monitoring of key performance indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

1.2 Purpose

The primary purpose of SECMS is to replace manual and fragmented enrollment processes with a scalable, automated, and secure Salesforce solution that ensures:

- Accurate student data management
- Structured enrollment workflows
- Automated approvals and validations
- Transparent reporting and monitoring

2. Ideation Phase

2.1 Problem Statement

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Enrollment Officer	manage student enrollments	enrollment data is	systems are manual and	frustrated and error-

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
		efficiently	scattered	unintegrated	prone
PS-2	Administrator	track enrollments and revenue	reporting is delayed	data is not centralized	uncertain in decision-making

2.2 Empathy Map Canvas

User Persona: Enrollment Officer

Dimension	Details
Think & Feel	Concerned about double bookings and revenue loss
Hear	Complaints from students and management
See	Multiple spreadsheets and emails
Say & Do	Rechecks data, follows up manually
Pain	Duplicate entries, approval bottlenecks
Gain	Automation, dashboards, faster processing

2.3 Brainstorming & Idea Prioritization

Step 1 – Brainstormed Ideas

- Spreadsheet automation
- Standalone enrollment portal
- ERP integration
- Salesforce-based enrollment system

Step 2 – Grouping

- Manual solutions

- Semi-automated solutions
- Fully automated cloud solutions

Step 3 – Prioritization

Salesforce-based SECMS was selected due to high feasibility and high impact.

3. Requirement Analysis Phase

3.1 Customer Journey Map

Stages covered:

- 1. Awareness & Registration
- 2. Course Selection
- 3. Enrollment Request
- 4. Fee Payment
- 5. Approval
- 6. Instructor Assignment
- 7. Reporting & Completion

3.2 Solution Requirements

Functional Requirements

FR No.	Epic	Description
FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve, reject enrollments
FR-4	Instructor Assignment	Auto-assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR-1: Usability

The system shall provide a simple, intuitive, and user-friendly interface that enables enrollment officers, administrators, instructors, and students to perform operations efficiently with minimal training. Navigation shall be organized through clearly labeled tabs, dashboards, and forms to reduce user effort and errors.

Examples: Quick student registration form; one-click enrollment approval; role-based dashboards

NFR-2: Security

The system shall ensure the confidentiality, integrity, and protection of student and institutional data through secure authentication, authorization, and access controls. Sensitive information shall only be accessible to authorized users.

Examples: Only admins can modify course details; instructors view only assigned students; students cannot access other student records

NFR-3: Reliability

The system shall operate consistently and accurately without failures during enrollment processing, approvals, and reporting. Automation logic shall execute correctly to avoid duplicate or incorrect enrollments.

Examples: No duplicate course registration; accurate status updates after approvals; consistent report generation

NFR-4: Performance

The system shall provide fast response times for all operations to ensure smooth and uninterrupted user experience, especially during peak admission periods.

Examples: Quick dashboard refresh; fast search for student records; no lag during bulk uploads

NFR-5: Scalability

The system shall support future growth in the number of students, courses, departments, and users without degradation in performance or functionality.

Examples: Add new departments without redesign; handle increased admissions annually; extend to mobile or portal integration

NFR-6: Availability

The system shall be continuously accessible to users during institutional working hours and must minimize downtime.

Examples: 24/7 cloud access; reliable during peak admission season; automatic backup of student records

3.3 Data Flow Diagram

- User inputs student and enrollment data
- System validates and stores records
- Approval process routes data
- Reports and dashboards consume processed data

The system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.

3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

4. Project Design Phase

4.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.

4.2 Proposed Solution

Parameter	Description
Problem	Manual enrollment handling

Parameter	Description
Solution	Salesforce-based SECMS
Novelty	End-to-end automation
Social Impact	Improved student experience
Revenue Model	Institutional deployment
Scalability	Multi-department support

4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers into a single Lightning application, ensuring that every stage of the enrollment lifecycle — from student registration through course completion — is captured and auditable within one connected data model.

5. Project Planning & Scheduling

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog & Sprint Schedule

Sprint	Epic	User Story #	User Story / Task	Pts	Priority	Owner
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create a Salesforce developer account so that I can configure and deploy the Student Enrollment system.	1	High	Member1

Sprint	Epic	User Story #	User Story / Task	Pts	Priority	Owner
Sprint-1	Data Modeling	USN-2	As an admin, I want to create custom objects for Student, Course, Enrollment, and Instructor so that academic data can be stored and managed efficiently.	3	High	Member2
Sprint-2	Data Modeling	USN-3	As a user, I want dedicated tabs for each object so that I can easily navigate between students, courses, enrollments, and instructors.	3	High	Member3
Sprint-2	Data Modeling	USN-4	As an admin, I want to create relevant fields such as student details, course information, and enrollment status so that complete records can be captured accurately.	5	High	Member4
Sprint-2	Data Modeling	USN-5	As an enrollment officer, I want a centralized Lightning App so that I can access all academic operations from one interface.	3	High	Member4
Sprint-2	Data Modeling	USN-6	As an admin, I want to define relationships between students, courses, and enrollments so that the system can logically link and track academic activities.	5	Medium	Member1
Sprint-2	Data Modeling	USN-7	As a user, I want customized page layouts so that only relevant information is displayed, improving clarity and ease of data	5	High	Member2

Sprint	Epic	User Story #	User Story / Task	Pts	Priority	Owner
			entry.			
Sprint-3	Automation	USN-8	As a system, I want to validate input data such as mandatory fields and enrollment limits so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-9	As a system, I want automated flows to handle enrollment processing and notifications so that manual effort is reduced and processes become faster.	3	High	Member4
Sprint-3	Automation	USN-10	As an administrator, I want an approval process for enrollment requests so that registrations are verified before confirmation.	5	High	Member1
Sprint-4	Security	USN-11	As an admin, I want to configure role-based security so that users can only access data relevant to their responsibilities.	5	High	Member2
Sprint-5	Reports	USN-12	As a manager, I want reports on student enrollments, course registrations, and instructor assignments so that I can monitor academic operations.	4	High	Member3
Sprint-6	Dashboards	USN-13	As a manager, I want visual dashboards showing enrollment statistics and trends so that I can	4	Medium	Member4

Sprint	Epic	User Story #	User Story / Task	Pts	Priority	Owner
			quickly analyze system performance.			

Project Tracker, Velocity & Burndown

Sprint	Total Pts	Duration	Start Date	Planned End	Pts Completed	Actual Release
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022	–	–
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	–	–
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022	–	–

6. Project Development Phase

Project Flow — Milestones

- Milestone 1: Salesforce Developer Account Creation
- Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)
- Milestone 3: Field Creation
- Milestone 4: Object Relationships
- Milestone 5: Validation Rules
- Milestone 6: Approval Process
- Milestone 7: Tabs Creation
- Milestone 8: Lightning App Creation
- Milestone 9: Page Layouts & Record Types
- Milestone 10: Flows (Automation)
- Milestone 11: Reports
- Milestone 12: Dashboards
- Milestone 13: Security Setup
- Milestone 14: Conclusion

What You'll Learn

- Real-time Salesforce project implementation
- Designing custom objects and relationships
- Creating page layouts & record types
- Writing validation rules
- Building automation with Flows
- Creating approval workflows
- Developing reports & dashboards
- Setting up security with profiles, permission sets, and sharing
- End-to-end application testing

Milestone 1: Salesforce Developer Account Creation

Introduction

Salesforce is a customer success platform designed to help organizations sell, service, market, analyze, and connect with their customers. Salesforce provides everything needed to run business operations from anywhere — managing relationships with prospects and customers, collaborating with employees and partners, and storing data securely in the cloud.

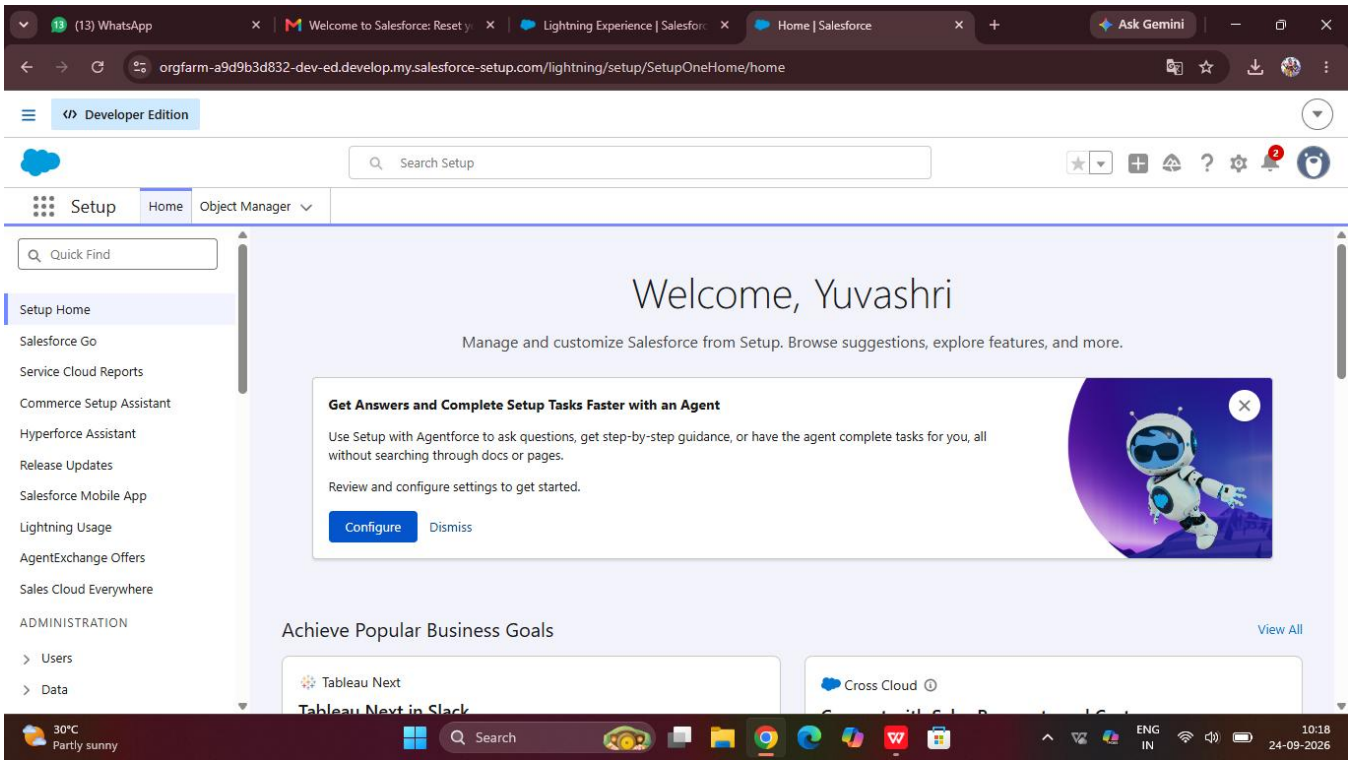
Activity 1: Creating a Developer Account

1. Go to https://developer.salesforce.com/signup
2. On the sign-up form, enter: First name & Last name, Email, Role (Developer), Company (College Name), Country (India), Postal Code
3. Set Username as a combination of your name and company (need not be a real email, format: username@organization.com)
4. Click Sign Me Up

Activity 2: Account Activation

1. Open the inbox of the email used to sign up and click Verify Account (may take 5–10 minutes)
2. Click Verify Account
3. Set a password and security question, then click Change Password

4. You will be redirected to the Salesforce Setup page



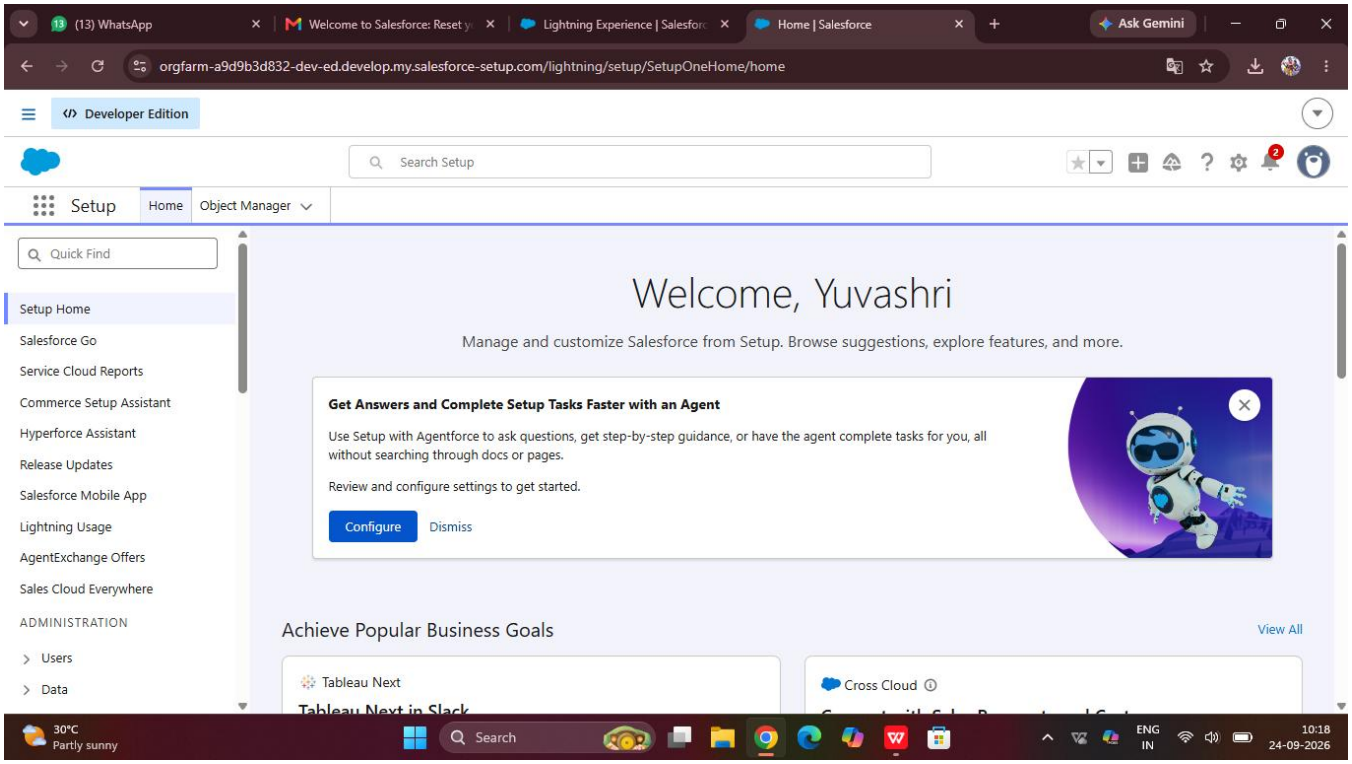
Milestone 2: Custom Object Creation

Activity 1: Creating the Student Object

The purpose of creating the Student custom object is to store and manage information about students such as their personal details, contact information, and enrollment status.

1. Go to the Setup page
2. Click Object Manager
3. Click Create → Custom Object
4. Enter Label: Student, Plural Label: Students, Object Name: Student, Record Name: Student Name (Data Type: Text)
5. Under Optional Features, select Allow Reports, Allow Activities, and Track Field History
6. Under Search Status, select Allow Search
7. Click Save

Repeat the same process to create the Course, Instructor, and Enrollment custom objects.



Milestone 3: Field Creation

Student Object Fields

Field Label	Data Type
Student Name	Text (Standard)
Email	Email
Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Course Object Fields

Field Label	Data Type
Course Name	Text (Standard)

Field Label	Data Type
Category	Picklist
Duration	Number / Text
Course Fee	Currency
Description	Text Area

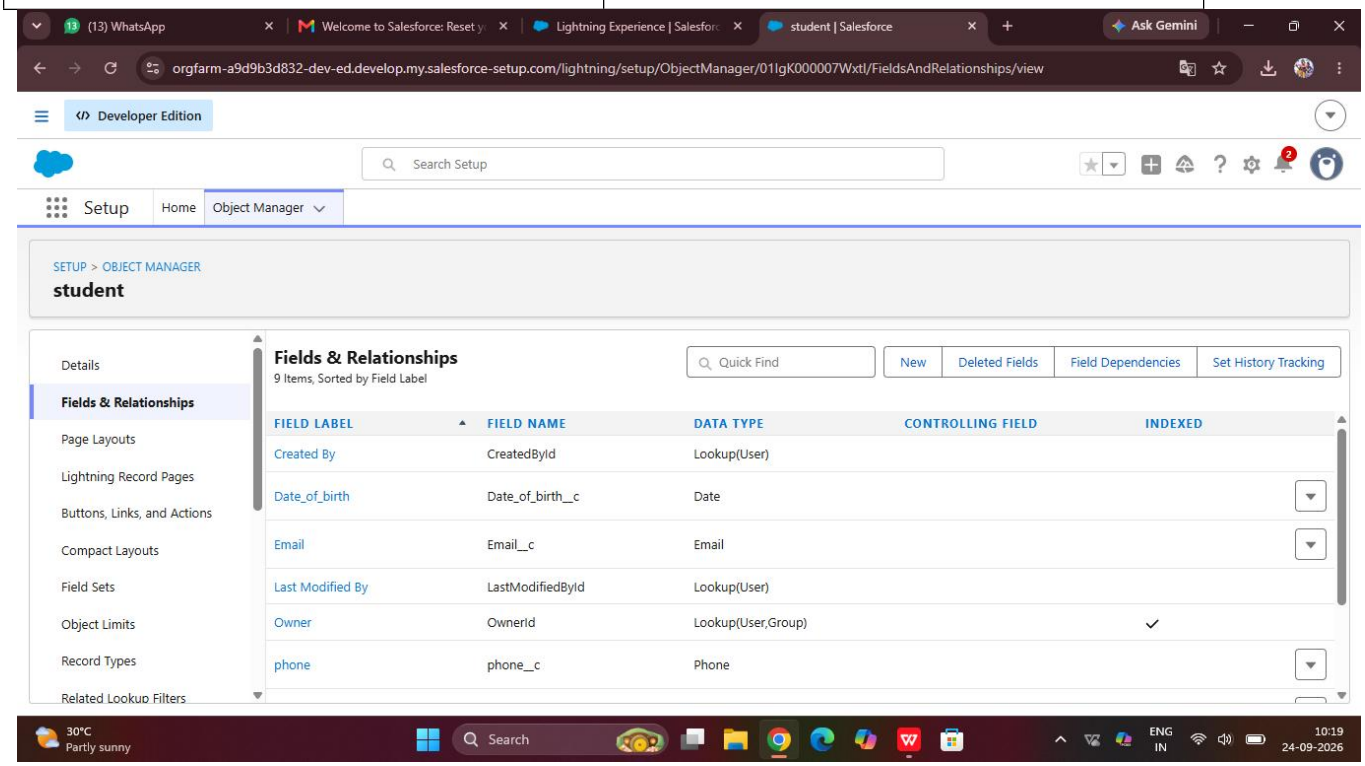
Instructor Object Fields

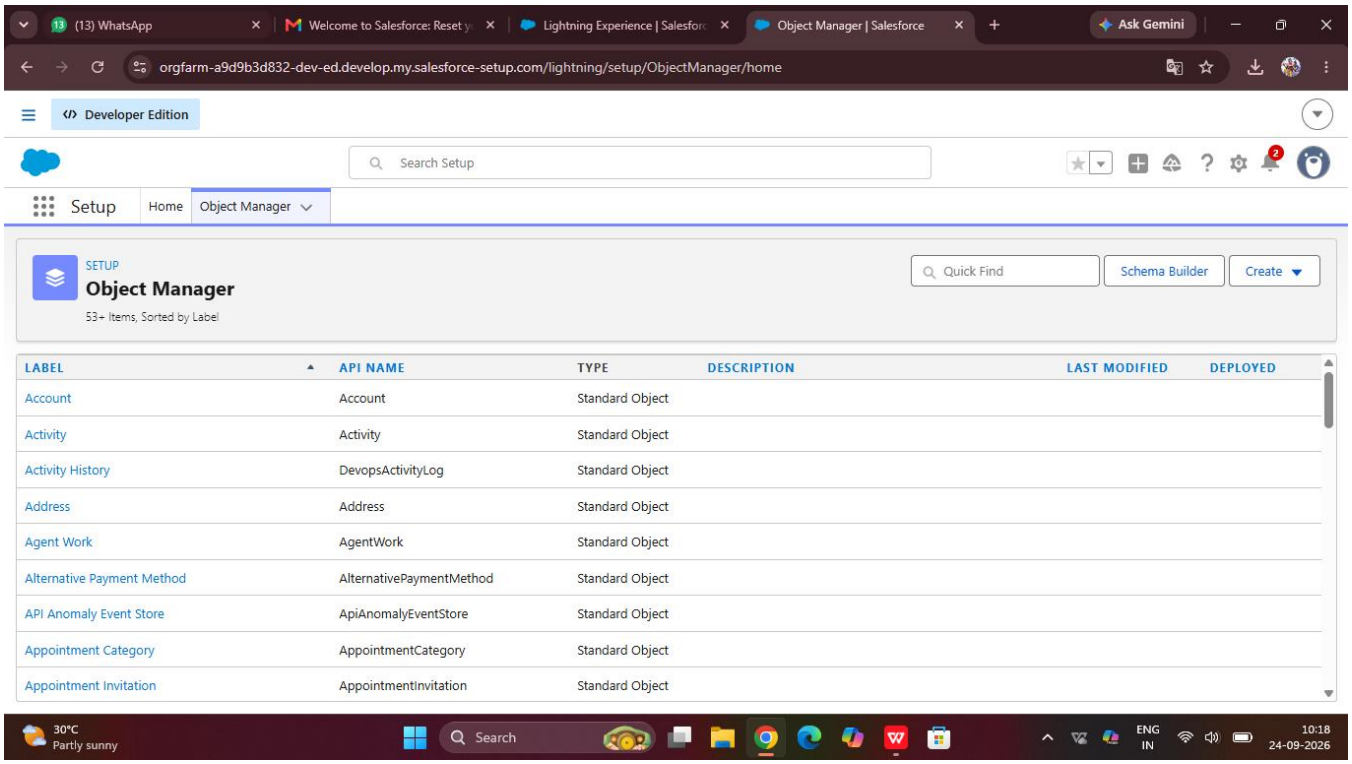
Field Label	Data Type
Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone
Email	Email

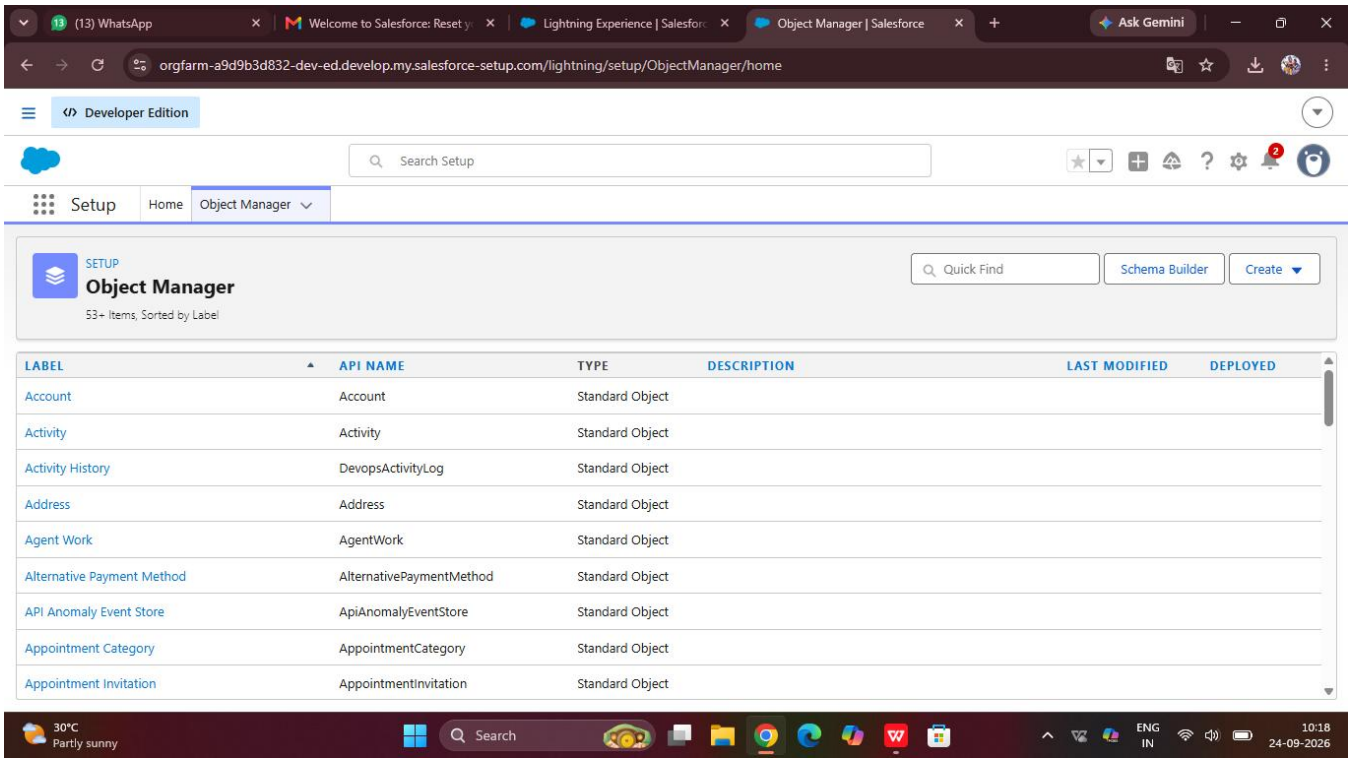
Enrollment Object Fields

Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Field Label	Data Type







Milestone 4: Create Relationships

Activity 1: Lookup from Enrollment to Student

1. Go to Setup → Object Manager
2. Type Enrollment in Quick Find → click the Enrollment__c object
3. Click Fields & Relationships → New
4. Select Lookup Relationship → Next
5. For Related To, select Student__c → Next
6. Enter Field Label: Student (API Name: Student__c)
7. Continue through Field-Level Security and Page Layout options, ensuring visibility
8. Click Save

Activity 2: Lookup from Enrollment to Course

Purpose: To link each Enrollment to a Course.

1. Go to Setup → Object Manager → Enrollment__c
2. Click Fields & Relationships → New

3. Select Lookup Relationship → Next
4. Related To: Course__c → Next
5. Field Label: Course (API Name: Course__c)
6. Set visibility and layout → Save

Activity 3: Lookup from Enrollment to Instructor

Purpose: To store which Instructor is assigned to a particular Enrollment.

1. Go to Setup → Object Manager → Enrollment__c
2. Click Fields & Relationships → New
3. Choose Lookup Relationship → Next
4. Related To: Instructor__c → Next
5. Field Label: Instructor (API Name: Instructor__c)
6. Complete Field-Level Security and Page Layout options → Save

Milestone 5: Validation Rules

Activity 1: Student Email Domain Validation

Purpose: To ensure students use only allowed institutional email addresses.

1. Go to Setup → Object Manager → Student__c
2. Click Validation Rules → New
3. Rule Name: Student_Email_Domain_Validation
4. Error Condition Formula: NOT(CONTAINS(Email__c, "@student.edu"))
5. Error Message: Email must be a valid student institutional email
6. Error Location: Field → Email__c
7. Click Save

Activity 2: Fees Must Be Paid Before Approval

Purpose: Prevent an Enrollment from being approved until fees are paid.

1. Go to Setup → Object Manager → Enrollment__c
2. Click Validation Rules → New
3. Rule Name: Fees_Must_Be_Paid_Before_Approval

4. Error Condition Formula: AND(ISPICKVAL(Enrollment_Status__c, "Approved"), Fees_Paid__c = FALSE)
5. Error Message: Fees must be paid before approving an enrollment
6. Error Location: Top of Page
7. Click Save

Milestone 6: Approval Process

Activity 1: Approval Process for Enrollment Requests

Purpose: Route Enrollment requests to the Training Manager for approval or rejection.

1. Go to Setup → Approval Processes → select Enrollment__c
2. Click Create New Approval Process → Use Standard Setup
3. Process Name: Approval_Enrollment_Request
4. Entry Criteria: Enrollment_Status__c = "Requested"
5. Save and configure steps
6. Add Approval Step — Approver: the designated Training Manager user
7. Under Final Approval Actions: add a Field Update to set Enrollment_Status__c = "Approved"
8. Under Final Rejection Actions: add a Field Update to set Enrollment_Status__c = "Rejected"
9. Add an Email Alert to notify the student of rejection
10. Activate the approval process

Milestone 7: Tabs Creation

Activity 1: Creating a Tab for the Student Object

1. Go to Setup
2. In the Quick Find box, type Tabs
3. Under Custom Object Tabs, click New
4. Select Student from the object list
5. Choose a tab style (icon)
6. Click Next, assign it to all profiles, then click Save

Activity 2: Creating Remaining Tabs

Repeat Activity 1 to create tabs for the Course, Instructor, and Enrollment objects.

Milestone 8: The Lightning App

Activity 1: Creating the Lightning Application

1. Navigate to Setup
2. Enter App Manager in the Quick Find search box
3. Click New Lightning App
4. App Name: Student Enrollment & Course Management (Developer Name auto-fills); add an optional description
5. Click Next and choose App Branding (optional logo/color theme) and Navigation Style: Standard Navigation
6. Click Next and assign the app to specific profiles (Admin, Enrollment Officer, Instructor)
7. Click Next and select items to include: Student, Course, Instructor, Enrollment, Reports, Dashboards, Home, Tasks
8. Click Next, review the app settings
9. Click Save & Finish

Milestone 9: Page Layouts & Record Types

Activity 1: Create Record Types for Enrollment

Purpose: To separate New Enrollment and Re-Enrollment processes.

1. Go to Setup → Object Manager → Enrollment__c
2. Click Record Types → New
3. First record type: Existing Record Type = Master, Record Type Label = New Enrollment; enable for Admin and Enrollment Officer profiles; select the default page layout → Save & New
4. Second record type: Record Type Label = Re-Enrollment; enable for the same profiles; select the default page layout → Save

Activity 2: Separate Page Layouts for Each Record Type

Purpose: To show different fields for New Enrollment versus Re-Enrollment.

1. Open Enrollment__c → click Page Layouts

2. Click New to create a layout for New Enrollment: clone from the existing Enrollment layout → name it New Enrollment Layout → Save
3. Keep fields relevant to first-time enrollment: Student, Course, Instructor, Enrollment Date, Status, Fees, etc.
4. Click New again to create the Re-Enrollment Layout: clone the existing layout → name it Re-Enrollment Layout → Save
5. Include prior-course history and re-enrollment-specific fields; remove fields not relevant to first-time sign-up
6. Click Save

Activity 3: Assign Layouts to Record Types

Purpose: To map each Record Type to its corresponding Page Layout, per profile.

1. Go to Enrollment__c → Record Types → select New Enrollment
2. Click Page Layout Assignment → Edit Assignment
3. For each profile, assign New Enrollment Layout → Save
4. Repeat for the Re-Enrollment record type, assigning Re-Enrollment Layout to each profile
5. Verify by creating a test record of each type and confirming the correct fields display

Milestone 10: Flows (Automation)

Activity 1: Auto-Assign Instructor Flow

Purpose: Automatically assign an available Instructor matching the Course category when an Enrollment is created.

1. Go to Setup → Flows → New Flow → select Record-Triggered Flow
2. Object: Enrollment__c; Trigger: A record is created
3. Add a Get Records element to find an Instructor__c whose Expertise matches the enrolled Course's Category
4. Add an Update Records element to set the Instructor__c lookup on the Enrollment record
5. Add a Decision element to handle the case where no matching instructor is found (create a Task for the Enrollment Officer instead)
6. Save as Auto_Assign_Instructor, activate the flow

Activity 2: Enrollment Notification Flow

Purpose: Notify the student and enrollment officer automatically as enrollment status changes.

1. Create a new Record-Triggered Flow on Enrollment__c, triggered when a record is updated
2. Entry condition: Enrollment_Status__c has changed
3. Add an Email Alert / Send Email action referencing the Student's email field
4. Add a Create Records element to log a Task for the Enrollment Officer on status change
5. Save as Enrollment_Status_Notification, activate the flow

Milestone 11: Reports

Purpose: Provide administrators and managers visibility into enrollment, course, and instructor activity.

1. Go to the Reports tab → New Report
2. Create a Tabular/Summary report type on Enrollments (with Student, Course, Instructor)
3. Report 1 – Enrollments by Status: group rows by Enrollment_Status__c, summarize Count
4. Report 2 – Course-wise Registrations: group by Course, summarize Total_Amount__c
5. Report 3 – Instructor Assignment Summary: group by Instructor, summarize Count of Enrollments
6. Add filters (e.g., Enrollment Date = This Fiscal Quarter) and save each report to a shared SECMS Reports folder

Milestone 12: Dashboards

Purpose: Visualize key enrollment metrics for quick decision-making.

1. Go to the Dashboards tab → New Dashboard, name it SECMS Enrollment Dashboard
2. Add a Donut chart component sourced from the Enrollments by Status report
3. Add a Bar chart component sourced from the Course-wise Registrations report
4. Add a Metric component showing total revenue (sum of Total_Amount__c where Fees_Paid__c = TRUE)
5. Add a Table component listing pending approvals (Enrollment_Status__c = Requested)
6. Set the running user and refresh schedule, then save and share the dashboard with relevant profiles

Milestone 13: Security Setup

Purpose: Ensure users can only access data relevant to their role.

1. Go to Setup → Profiles: review and restrict object/field permissions per role (Admin, Enrollment Officer, Instructor)
2. Create Permission Sets for elevated access where needed (e.g., approving enrollments) instead of broadening base profiles
3. Set Organization-Wide Defaults for Enrollment__c to Private, then use Sharing Rules so Instructors see only their assigned students
4. Configure Role Hierarchy so managers can view records owned by their subordinates
5. Enable Field-Level Security so sensitive fields (e.g., fee/payment data) are hidden from the Instructor profile
6. Test each profile by logging in as a test user and confirming visibility matches expectations

Milestone 14: Conclusion

The Student Enrollment & Course Management System (SECMS) successfully demonstrates how Salesforce's native capabilities — custom objects, relationships, validation rules, Flows, Approval Processes, Reports, and Dashboards — can be combined to replace a fragmented, manual enrollment process with a single, automated, and secure platform.

Through this project, custom objects and relationships were designed to model the academic domain, automation was built to reduce manual effort in instructor assignment and notifications, approval workflows were configured to enforce accountability, and role-based security was applied to protect sensitive student data. The resulting system gives enrollment officers, administrators, and instructors real-time visibility into enrollment volume, revenue, and pending approvals — fulfilling the project's core goal of centralized, transparent, and reliable academic operations management.